

ECE131a Extra Credit Project

Distributed Detection in Gaussian Noise

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1 Introduction

1.1 Background

This problem is adapted from the results in [1] and [2]. A scalar signal $S \in \{-m, m\}$ is transmitted from a satellite to a field of n sensors on the surface of the earth. S communicates an important message, and is equally likely to take either value so that

$$P_S(s) = \begin{cases} \frac{1}{2}, & \text{for } s = -m, \\ \frac{1}{2}, & \text{for } s = m, \\ 0, & \text{otherwise.} \end{cases}$$

Sensor i observes random variable

$$X_i = S + Z_i$$

where the random variables Z_i are independent and identically distributed according to a standard normal distribution $\mathcal{N}(0, 1)$. This project explores how to efficiently and reliably estimate S from the vector of observations

$$\mathbf{X} = [X_1, X_2, \dots, X_n].$$

Any estimate $\hat{S}$ will naturally depend on the observables. A *statistic* is a function of the set of observables that we use in preparing our estimate. For this problem we will be especially interested in the statistic

$$T_n = \frac{1}{n} \sum_{i=1}^n X_i,$$

which is sometimes called the sample mean.

A *sufficient statistic* of a set of observations for estimation of a particular random variable is one that allows as good of an estimate of the random variable as the original observations themselves. T_n is a sufficient statistic of $\mathbf{X}$ for S if and only if

$$f_{X|T,S}(x|t, s) = f_{X|T}(x|t).$$

In our specific case, this turns out to be true so that the sample mean is a sufficient statistic of $\mathbf{X}$ for S . We can use the sample mean T_n to estimate S and be confident that we have not lost any information. Specifically, we will estimate S as follows:

$$\hat{S} = \begin{cases} m, & \text{if } T_n \geq 0, \\ -m, & \text{otherwise.} \end{cases}$$

1.2 Report Organization

The project is 25 project points which is equivalent to 5% of the total grade percentage as extra credit. The project must be completed to earn an A+ grade.

There are 8 problems, each answer must have a Matlab plot to back up the answer.

Project Statement:

ECE 131a Spring 2026 Extra Credit Project, Distributed Detection in Gaussian Noise

2 Problem 1

3pts.

Suppose that $m = 0.5$. Use Matlab function `semilogy` to plot the probability of error $P(\hat{S} \neq S)$ as a function of the number of sensors n for $1 \leq n \leq 100$. The Matlab function `qfunc` is your friend for this problem. How many sensors do you need so that $P(\hat{S} \neq S) < 10^{-3}$? Call this number n_g (as in the number of Gaussian samples).

For this problem, each sensor observes

$$X_i = S + Z_i$$

where $S \in \{-m, m\}$ and the noise variables Z_i are independent and identically distributed according to a standard normal distribution $N(0, 1)$. The signal amplitude is given as $m = 0.5$.

The detector uses the sample mean

$$T_n = \frac{1}{n} \sum_{i=1}^n X_i$$

and makes the decision

$$\hat{S} = \begin{cases} m, & T_n \geq 0 \\ -m, & T_n < 0 \end{cases}$$

The goal is to determine the probability of error as a function of the number of sensors n and find the minimum number of sensors required so that the probability of error is less than 10^{-3} .

2.1 Distribution of the Sample Mean

Starting with the definition of the sample mean,

$$T_n = \frac{1}{n} \sum_{i=1}^n (S + Z_i)$$

Since S is constant across all sensors,

$$T_n = S + \frac{1}{n} \sum_{i=1}^n Z_i$$

Because the noise variables Z_i are independent Gaussian random variables with mean 0 and variance 1, the average noise term has distribution

$$\frac{1}{n} \sum_{i=1}^n Z_i \sim N\left(0, \frac{1}{n}\right)$$

Therefore, conditioned on the transmitted signal S ,

$$T_n|S = m \sim N\left(m, \frac{1}{n}\right)$$

and

$$T_n|S = -m \sim N\left(-m, \frac{1}{n}\right)$$

Thus, the sample mean remains to be Gaussian with mean equal to the transmitted signal (m) and variance equal to $1/n$.

2.2 Probability of Error

An error occurs when the detector chooses the wrong hypothesis.

Considering the case $S = m$, the detector makes an error whenever $T_n < 0$. Therefore,

$$P(\text{error}|S = m) = P(T_n < 0|S = m)$$

Since $T_n|S = m$ is Gaussian with mean m and variance $1/n$, we are able to standardize the random variable:

$$\begin{aligned} P(\text{error}|S = m) &= P\left(\frac{T_n - m}{1/\sqrt{n}} < \frac{0 - m}{1/\sqrt{n}}\right) \\ &= P(Z < -m\sqrt{n}) \end{aligned}$$

where $Z \sim N(0, 1)$.

Using the Gaussian Q-function,

$$P(\text{error}|S = m) = Q(m\sqrt{n})$$

By symmetry,

$$P(\text{error}|S = -m) = Q(m\sqrt{n})$$

Since both of the hypotheses are equally likely,

$$\begin{aligned} P(\text{error}) &= \frac{1}{2}P(\text{error}|S = m) + \frac{1}{2}P(\text{error}|S = -m) \\ &= Q(m\sqrt{n}) \end{aligned}$$

Substituting $m = 0.5$:

$$P(\text{error}) = Q(0.5\sqrt{n})$$

This expression above is the theoretical probability of error for any number of sensors n .

2.3 MATLAB

The probability of error was evaluated for $n = 1$ to 100 using MATLAB's `qfunc` function. A semilogarithmic plot was generated using `semilogy` to clearly display the exponential decrease in error probability as the number of sensors increases.

The MATLAB results show that the probability of error decreases rapidly as additional sensors are added. This occurs because averaging more independent observations reduces the variance of the sample mean, making it easier to distinguish between the two possible signal values.

The smallest number of sensors satisfying

$$P(\hat{S} \neq S) < 10^{-3}$$

was found to be

$$n_g = 39.$$

Therefore, at least 39 sensors are required to achieve a probability of error below 10^{-3} .

Listing 1: Matlab Code for Problem 1

```
1 clc;
2 clear;
3 close all;
4
5 %% Problem 1
6 % Distributed Detection in Gaussian Noise
7 % P(error) = Q(m*sqrt(n))
8
9 m = 0.5;           % Signal amplitude
10 n = 1:100;         % Number of sensors
11
12 % Compute probability of error
13 Pe = qfunc(m*sqrt(n));
14
15 % Error threshold
16 threshold = 1e-3;
17
18 % Find smallest n satisfying Pe < 10^-3
19 ng = find(Pe < threshold, 1, 'first');
20
21 %% Create Figure
22 figure;
23
24 semilogy(n, Pe, 'LineWidth', 2);
25 hold on;
26 grid on;
27
28 % Threshold line
29 yline(threshold, '--', '10^{-3}', 'LineWidth', 1.5);
30
31 % Mark ng
32 semilogy(ng, Pe(ng), 'o', ...
```

```

33     'MarkerSize', 8, ...
34     'LineWidth', 2);
35
36 xlabel('Number of Sensors, n');
37 ylabel('Probability of Error');
38 title('Probability of Error vs. Number of Sensors (m = 0.5)');
39
40 legend('P_e = Q(0.5\sqrt{n})', ...
41        'Threshold', ...
42        sprintf('n_g = %d',ng), ...
43        'Location','southwest');
44
45 %% Display Results
46 fprintf('\n');
47 fprintf('-----\n');
48 fprintf('Problem 1 Results\n');
49 fprintf('-----\n');
50 fprintf('Signal amplitude m = %.1f\n',m);
51 fprintf('Required error probability < 10^-3\n');
52 fprintf('Minimum number of sensors n_g = %d\n',ng);
53 fprintf('Probability of error at n_g = %.6e\n',Pe(ng));
54
55 if ng > 1
56     fprintf('Probability of error at n = %d is %.6e\n', ...
57            ng-1, Pe(ng-1));
58 end
59 fprintf('-----\n');

```

3 Problem 2

4pts.

Simulate this sensor system with the n_g sensors as follows: For each simulation run, generate n_g observations X_i using $S = 0.5$ and `normrnd` to generate Z_i and use them to compute the sufficient statistic T_{n_g} . Then use T_{n_g} to compute $\hat{S}$ and note whether the estimate is correct or is an error. Perform simulation runs until you have observed 200 errors. Keep track of the total number K of simulation runs. Report your estimated error rate as $200/K$. If everything is working well, your simulation point should match the probability of error you calculated in the previous part for n_g observations, i.e., a number close to 10^{-3} . Your simulation may take a few minutes to complete since you will need approximately 200,000 runs.

4 Problem 3

3pts.

We will refer to the sensor studied in problems 1–2 as a “full precision” sensor. Now, let’s suppose that it is resource intensive to sense the real numbers X_i with full precision. We want to deploy less expensive sensors that merely detect whether $X_i \geq 0$. These sensors each collect

$$B_i = \begin{cases} 1 & \text{if } X_i \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

We are fortunate that $U_n = \sum_{i=1}^n B_i$ is a sufficient statistic of B for S , so that we can base our estimate $\hat{S}$ on this sum.

What kind of random variable is B_i ? What kind of random variable is U_n ? For our case of interest, where $S = \pm 0.5$ and $X_i = S + Z_i$ with the random variables Z_i independent and identically distributed according to a standard normal distribution $\mathcal{N}(0, 1)$, find the value of $P(B_i = 1 \mid S = 0.5)$, which we will call p .

We need a decision rule to decide our estimate $\hat{S}$ as a function of U_n . We will use the “maximum likelihood” principle to choose our rule. Specifically, choose the value of $\hat{S}$ the rule that maximizes the probability of the observed value of U_n given $S = \hat{S}$.

$$\hat{S} = \begin{cases} 0.5 & \text{if } P_{U|S}(u|0.5) > P_{U|S}(u|-0.5) \\ -0.5 & \text{otherwise.} \end{cases}$$

Show that the maximum-likelihood decoder turns out to be a majority vote decoder. It turns out that how you handle ties does not affect the total probability of error, but it could favor one of the outcomes.

5 Problem 4

3pts.

Use the Matlab function `semilogy` to plot the probability of error $P(\hat{S} \neq S)$ as a function of the number of sensors n for $1 \leq n \leq 100$ for the system based on sensors that provide B_i . Add this to your plot from problem 2 so that you have a single plot with two curves. Label your axes. Include a legend. Take pride in your data. For the binary sensor, you only need to plot probability of error for odd values of n , so you don't need to worry about breaking ties. The Matlab function `binocdf` is your friend for this problem. How many sensors, considering only odd numbers, do you need so that $P(\hat{S} \neq S) < 10^{-3}$? Call this number n_b (as in the number of Bernoulli samples in the binomial random variable.)

6 Problem 5

3pts.

Simulate this binary sensor system with n_b sensors as follows: For each simulation run, generate n_b observations X_i as before and then compute the associated B_i values and use them to compute the sufficient statistic U_{n_b} . Then use U_{n_b} to compute $\hat{S}$ and note whether the estimate is correct or is an error. Perform simulation runs until you observe 200 errors. Keep track of the total number K of simulation runs. Report your estimated error rate as $\frac{200}{K}$. If everything is working well, your simulation point should match the probability of error you calculated in the previous part for n_b observations, i.e. a number close to 10^{-3} .

7 Problem 6

3pts.

If the binary sensor that we studied in problems 3–5 costs half as much as the full-precision sensor we studied in problems 1–2, which system costs less to achieve probability of error 10^{-3} ? Based on the slopes of the lines in your plot, would the same system also be the most cost effective for lower probabilities of error, assuming binary sensors costs half as much as the full-precision sensors?

8 Problem 7

3pts.

As it turns out, the binary sensors as we have defined them above have a peculiar property that for an odd number n , the probability of error for n sensors is the same as the probability of error for $n + 1$ sensors. Confirm this with a few calculations using `binocdf`. You can break ties, for example, by always calling them $\hat{s} = m$. Show mathematically that the probability of error for n sensors is the same as the probability of error for $n + 1$ sensors for any odd number n and any value of p .

9 Problem 8

3pts.

Devise a modification to the sensor for the even-valued n case that will allow it to perform better than the case with one fewer sensor. If you can't think of a way, look at Fig. 7 in [1] and its discussion for a clue. For the value n_b that you identified in problem 4, optimize your modification to get the most improvement possible in the probability of error. It won't be too much better. You will never do better than $n_b + 2$ sensors using the standard approach we described above. Report the probability of error you obtain by applying your modification for the case of $n_b + 1$ sensors.

10 Conclusion